

KAZI TANJIM SHAKIB

Senior Software Engineer | iOS Developer | Applied AI Builder

Dhaka, Bangladesh | 4+ years of software engineering experience

Production-minded iOS engineer focused on reliable mobile experiences, maintainable architecture, performance engineering, and privacy-conscious AI applications.

Current role	Senior Software Engineer, Samsung R&D Institute Bangladesh
Primary specialization	Swift, UIKit, SwiftUI, iOS architecture, API integration, testing and performance
Core product experience	SmartThings iOS application
Education	B.Sc. in Computer Science & Engineering, Rajshahi University of Engineering & Technology (RUET)
GitHub	github.com/Shakib053
Professional email	kazitanjimshakib053@gmail.com

This profile presents professional experience, technical knowledge, project work, achievements, and ongoing learning in a clear, searchable structure.

1. Executive Professional Summary

Kazi Tanjim Shakib is a Senior Software Engineer and iOS developer based in Dhaka, Bangladesh. He has worked at Samsung R&D Institute Bangladesh since August 2021 and has progressed from Internship Trainee to Software Engineer and then Senior Software Engineer.

His professional work centers on production iOS development using Swift, UIKit and SwiftUI. He contributes to the SmartThings iOS application through feature implementation, module ownership, device and service plugin API integration, debugging, refactoring, automated testing, performance analysis, code review and user-experience improvements.

Alongside mobile development, he is building practical knowledge in on-device artificial intelligence, local large language models, embeddings, semantic search, keyword search, hybrid retrieval, reranking, retrieval-augmented generation and agentic workflows. His goal is to combine strong iOS engineering with privacy-aware AI features that can run locally or use carefully designed cloud fallbacks.

Professional Value Proposition

- More than four years of professional software engineering experience, beginning in August 2021.
- Hands-on experience developing and maintaining features in a globally distributed consumer iOS application.
- Strong working knowledge of Swift, UIKit, SwiftUI, application architecture, networking, concurrency, testing and performance engineering.
- Recognized for code-review quality, individual contribution, engineering excellence and team delivery.
- Practical project experience with MLX Swift, local LLMs, Hugging Face models, RAG pipelines and retrieval systems.
- Able to explain engineering decisions clearly and connect product requirements with architecture, quality and long-term maintainability.

Career Snapshot

Current position	Senior Software Engineer at Samsung R&D Institute Bangladesh
Professional domain	iOS application development and software engineering
Experience period	August 2021 to present
Primary programming language	Swift
Additional languages	Python, C and C++
Working languages	Bengali (native) and English (full professional proficiency)

2. Professional Experience

Samsung R&D Institute Bangladesh

Location: Dhaka, Bangladesh

Employment period: August 2021 - Present

Career progression: Internship Trainee -> Software Engineer -> Senior Software Engineer

Role	Period
Senior Software Engineer	March 2024 - Present
Software Engineer	February 2022 - February 2024
Internship Trainee	August 2021 - January 2022

Core Responsibilities and Contributions

- Develop and maintain iOS features for the SmartThings application using Swift and Apple platform frameworks.
- Own and support multiple application modules across feature delivery, defect resolution, maintenance and refactoring.
- Integrate device-plugin and service-plugin APIs that connect application flows with SmartThings devices and services.
- Investigate crashes, functional defects, stability issues and usability problems in a large production codebase.
- Review code and provide actionable feedback on correctness, readability, maintainability, architecture and risk.
- Develop unit tests and UI tests and apply test-driven development principles where they provide practical value.
- Use Xcode Instruments and diagnostics to investigate memory usage, hangs, hitches and runtime responsiveness.
- Collaborate through Git, JIRA, Agile and Scrum practices during development, validation and release activities.
- Create proof-of-concept implementations to validate technical approaches and emerging product ideas.
- Contribute to product stability and App Store quality through systematic debugging, regression prevention and performance work.

Selected Production Engineering Work

- SmartThings iOS: contributed to a globally distributed consumer application used at large scale.
- API integration ownership: supported flows from device cards to plugins and validated multiple device and service plugin interfaces.
- Performance automation: built Python and Bash automation around Apple Instruments to execute UI scenarios on physical devices and collect performance reports.
- Architecture and refactoring: worked with MVC, MVVM and MVVM-C patterns across UIKit and SwiftUI codebases.
- On-device AI proofs of concept: explored Siri-assisted device control, voice-based error reporting and privacy-oriented generative experiences using Apple platform capabilities.

3. Recognition and Professional Achievements

Best Code Reviewer - Q4 2025

Recognized for high-quality code review, careful engineering feedback and contribution to codebase quality. The recognition was noted as a top-tier organizational achievement.

Icon of the Month - March 2024

Recognized for strong individual performance and contribution.

Team of the Year - 2023

Member of a team recognized for exceptional collective delivery and collaboration.

Excellence Award - Q4 2022

Received an organizational excellence award early in his engineering career.

LeetCode consistency milestones - 2024

Earned 50-day and 100-day problem-solving badges through sustained algorithm and data-structure practice.

Professional Strengths Demonstrated by These Achievements

- Attention to engineering quality and maintainability.
- Consistency in learning and problem solving.
- Ability to collaborate effectively within a team.
- Willingness to take ownership of difficult technical work.
- Strong communication through code review and technical feedback.

4. iOS and Software Engineering Competencies

iOS and Apple Platforms

Swift; UIKit; SwiftUI; Foundation; Combine; ReactiveSwift; MapKit; WidgetKit; WatchKit; Core Data; SwiftData; Firebase integration; App Store delivery concepts.

Architecture and Design

MVC; MVVM; MVVM-C; Coordinator and Router patterns; protocol-oriented design; state management; separation of concerns; dependency injection concepts; SOLID principles; Clean Code; modular design.

Networking and Data

URLSession; REST API integration; HTTP methods and headers; JSON encoding and decoding; authentication integration; error modeling; retry and fallback strategies; caching concepts; persistence and data flow.

Concurrency and Reactive Programming

Swift async/await; Task; async let; structured concurrency; actors; MainActor; Grand Central Dispatch; Combine publishers and subscribers; cancellable lifecycle management; ReactiveSwift.

Testing and Quality

Unit testing; UI testing; test-driven development; code review; debugging; regression prevention; testability through dependency separation; production issue investigation.

Performance and Tooling

Xcode Instruments; memory analysis; hangs and hitches; responsiveness diagnostics; Python and Bash automation; Git; JIRA; Postman; Xcode debugging and profiling tools.

General Software Engineering

Requirement analysis; API contract understanding; modular implementation; technical trade-off analysis; documentation; refactoring; maintainability; release readiness; collaboration in Agile teams.

Engineering Principles

- Prefer simple, maintainable abstractions over unnecessary complexity.
- Separate UI, navigation, business logic, networking and persistence responsibilities.
- Treat testing, observability, performance and error handling as part of feature development.
- Use protocol-driven boundaries and dependency injection where they improve testability and flexibility.
- Choose on-device processing when privacy, latency, offline use or cost makes it valuable.
- Build learning projects as complete systems with documentation, architecture, error handling and measurable user value.

5. Applied AI, LLM and Retrieval Knowledge

Kazi is developing applied AI skills from a software-engineering perspective. His focus is not only calling a model API, but understanding how data is prepared, retrieved, ranked, passed to a model and presented safely in a user-facing application.

Large Language Model Foundations

Tokenization; next-token prediction; context windows; temperature; top-k and top-p sampling; prompt construction; hallucination risks; response post-processing; hosted and local inference.

Document Processing

Loading TXT and PDF files; preserving useful metadata; cleaning text; selecting chunk size and overlap; recursive and semantic chunking concepts; designing chunks that remain meaningful after retrieval.

Embeddings and Semantic Search

Creating vector representations of text with sentence-transformer models; comparing query and document vectors; retrieving conceptually related passages even when wording is different; understanding similarity scores and embedding-model trade-offs.

Keyword and Hybrid Retrieval

Using BM25 for exact terms, identifiers and uncommon keywords; combining BM25 with dense vector search; balancing retriever weights; using reciprocal-rank or ensemble approaches to improve recall.

Reranking

Using a cross-encoder to jointly evaluate a query and candidate passage; selecting the strongest final context after broad first-stage retrieval; balancing relevance improvement with latency and compute cost.

Grounded Generation

Constructing prompts that instruct the language model to answer from retrieved evidence; handling missing evidence; preserving source metadata; reducing unsupported claims; maintaining conversation history without overwhelming the context window.

Local and On-Device AI

Using Ollama for local model experimentation and MLX Swift for Apple-device inference; understanding simulator versus physical-device constraints; designing local-first systems with hosted fallbacks.

Agentic and Workflow Systems

Learning linear workflows, tool-calling patterns, state transitions, routing, role-based agents and production orchestration through frameworks such as LangChain, LangGraph and LlamaIndex.

6. Selected Portfolio Projects

MLX-SwiftUI - On-Device AI Chat

Technology: SwiftUI, MLX Swift, MLXLMCommon, MLXLLM, MLXHuggingFace, Hugging Face API

An iOS project for experimenting with local large language models and privacy-conscious AI experiences on Apple devices.

- Runs supported language models locally on compatible physical devices through MLX-based tooling.
- Uses a simulator-aware fallback design because local Metal-backed execution can behave differently in the iOS simulator.
- Explores a shared chat-backend abstraction so local MLX inference and hosted Hugging Face inference can be swapped cleanly.
- Includes response filtering so model-specific reasoning tags are not displayed in the user-facing chat UI.
- Planned direction includes selectable models, local document retrieval, offline utilities and production-ready error states.

Basic RAG System - Local Retrieval-Augmented Generation

Technology: Python, LangChain, ChromaDB, sentence-transformers, BM25, CrossEncoder, Ollama

A document-question-answering project designed to retrieve relevant evidence before generating an answer.

- Loads TXT and PDF documents, normalizes their content and splits them into searchable chunks.
- Creates dense embeddings using a Hugging Face sentence-transformer model and stores vectors in ChromaDB.
- Combines semantic retrieval with BM25 keyword retrieval to improve both conceptual and exact-term matching.
- Uses ensemble retrieval and cross-encoder reranking to improve final context relevance.
- Integrates local language models through Ollama and supports conversation-history experiments.
- Provides a practical platform for testing chunking, metadata, retrieval quality, reranking and grounded prompting.

Stock Tracker Website - Dhaka Stock Exchange Portfolio Tool

Technology: JavaScript, Netlify Functions, DSE quote parsing, web UI

A personal web application for tracking Dhaka Stock Exchange holdings, transactions and market prices.

- Tracks buy and sell transactions, cost basis and portfolio performance.
- Uses serverless functions and parsing logic to retrieve market data.
- Includes investigation of production deployment, module-format, caching and stale-data issues.
- Planned enhancements include specific-lot selling, Bangladesh fiscal-year reporting and a professional dark trading-terminal interface.

Money Manager iOS

Technology: SwiftUI, SwiftData, MVVM-C, export and notification concepts

A personal finance application concept focused on clean SwiftUI architecture and useful money-management workflows.

- Explores transaction management, persistence, settings architecture and coordinator-based navigation.
- Planned capabilities include CSV/PDF export, currency conversion, configurable notifications and modular feature design.

Developer Productivity and Performance Automation

Technology: Python, Bash, Xcode UI testing, Instruments

Automation that executes UI flows on a physical iOS device and collects performance diagnostics.

- Reduces repetitive manual testing and improves consistency of memory and responsiveness measurements.
- Demonstrates the use of scripting to extend mobile-development and quality-engineering workflows.

7. Learning Journey and Current Development Areas

Swift and iOS Depth

- Structured concurrency, actors, MainActor and cancellation.
- UIKit and SwiftUI integration patterns.
- MVVM-C, Coordinator and Router navigation.
- URLSession networking layers, CRUD operations and error handling.
- WidgetKit timelines and refresh behavior.
- Performance profiling, memory optimization and responsive interfaces.

AI and Retrieval Systems

- LLM fundamentals and generation controls.
- Embeddings and vector databases.
- Semantic chunking and retrieval evaluation.
- BM25, hybrid search and reranking.
- LangChain, LangGraph, LlamaIndex and linear workflow agents.
- Local models through Ollama and on-device inference through MLX Swift.

Production Engineering

- CI/CD concepts and release automation.
- Improved observability and diagnostic tooling.
- Clearer architecture boundaries and dependency management.
- Security-conscious handling of API tokens and configuration.
- Documentation, README quality and professional project presentation.

Learning Approach

His preferred learning method is implementation-heavy: learn the core concept, build a small working version, integrate it into a realistic project, test edge cases, document architecture and trade-offs, and then present the result through GitHub, a portfolio demonstration or a technical discussion.

8. Education and Professional Identity

Bachelor of Science in Computer Science & Engineering

Institution: Rajshahi University of Engineering & Technology (RUET), Bangladesh

The degree established a foundation in programming, data structures, algorithms, object-oriented design, computer systems, databases, networking and software engineering. This academic foundation supports his professional mobile-development work and his continued study of artificial intelligence and retrieval systems.

Professional Identity

- Senior Software Engineer with a primary specialization in iOS development.
- Swift developer experienced with both UIKit and SwiftUI.
- Production engineer with module ownership, API integration, testing, debugging and performance responsibilities.
- Applied AI learner building on-device AI and retrieval-augmented generation projects.
- Engineer who values reliability, maintainability, clear communication and practical user value.

Languages and Collaboration

Bengali is his native language. He uses English with full professional proficiency for technical communication, documentation, code review and collaboration.

9. Professional Knowledge Index

The following concise statements summarize key facts using common terms that may appear in technical discussions, recruiter questions, search queries or project demonstrations.

Who is Kazi Tanjim Shakib?

Kazi Tanjim Shakib is a Senior Software Engineer and iOS developer based in Dhaka, Bangladesh.

Where does he work?

He works at Samsung R&D Institute Bangladesh and has been employed there since August 2021.

What is his current role?

His current role is Senior Software Engineer.

How much professional experience does he have?

He has more than four years of professional software engineering experience, starting in August 2021.

What product has he worked on?

His primary production product experience is the SmartThings iOS application.

What are his main iOS technologies?

His main iOS technologies are Swift, UIKit, SwiftUI, Foundation, Combine, ReactiveSwift, Core Data and SwiftData.

What architecture patterns does he know?

He has working knowledge of MVC, MVVM, MVVM-C, Coordinator and Router patterns, SOLID principles and separation of concerns.

What networking experience does he have?

He works with URLSession, REST APIs, JSON, HTTP methods, headers, authentication, error handling and API integration.

What testing experience does he have?

He has experience with unit testing, UI testing, test-driven development, debugging, regression prevention and code review.

What performance experience does he have?

He uses Xcode Instruments to investigate memory usage, hangs, hitches and runtime responsiveness, and has built Python and Bash automation for performance testing.

What AI and RAG knowledge does he have?

He has hands-on learning and project experience with embeddings, semantic search, BM25, hybrid retrieval, ChromaDB, cross-encoder reranking, Ollama, Hugging Face, LangChain and MLX Swift.

What is his RAG project?

His Basic RAG System loads text and PDF documents, creates embeddings, stores them in ChromaDB, combines semantic search with BM25 and reranks retrieved passages before local LLM generation.

What is his on-device AI project?

MLX-SwiftUI is an iOS chat project that explores local language-model inference on Apple devices with a hosted inference fallback for unsupported environments.

What awards has he received?

His recognitions include Best Code Reviewer Q4 2025, Icon of the Month March 2024, Team of the Year 2023 and Excellence Award Q4 2022.

Where did he study?

He completed a B.Sc. in Computer Science & Engineering at Rajshahi University of Engineering & Technology, commonly known as RUET.

What roles is he suited for?

He is suited for mid-level to senior iOS engineering roles, especially roles involving Swift, UIKit, SwiftUI, architecture, API integration, quality, performance or on-device AI.

10. Topics He Can Discuss in a Technical Interview

- How to design a testable networking layer in Swift using URLSession, protocols and dependency injection.
- How MVVM-C separates screen state, business logic and navigation in UIKit or SwiftUI applications.
- How structured concurrency, actors and MainActor help manage asynchronous work and UI safety.
- How to investigate memory growth, hangs and hitches using Xcode Instruments.
- How to design a local-first AI application with a clean fallback to hosted inference.
- How document chunking, embeddings and metadata affect retrieval quality.
- Why semantic search and BM25 solve different retrieval problems and how hybrid search combines them.
- How cross-encoder reranking improves context quality after first-stage retrieval.
- How to prevent model-specific reasoning tags or internal formatting from appearing in a production chat interface.
- How to turn a learning project into a presentable engineering portfolio project through documentation, architecture and measurable behavior.

Closing Profile

Kazi Tanjim Shakib brings together professional iOS product development, disciplined software-engineering practices and a growing applied-AI skill set. His strongest differentiator is the ability to connect production mobile engineering with practical, privacy-aware AI and retrieval experiences while continuing to strengthen architecture, performance, testing and system-design knowledge.